Trøndelagkonkurranse 2026 - Level II

Third Round - October 31, 2026

2nd Part

Frida wrote the following sequence of numbers

$$011000111100000111111\ldots$$

starting with 0, then two 1's, then three 0's, then four 1's and so on. Skye then asked Frida: 'How many numbers are you going to write?'. To which she replied, 'Just 2026 numbers'.

Task 1: Determine if the number in the 2026th position is a 0 or a 1.

Skye, thrilled, started doing something similar, but instead he wrote the following sequence

$$112123123412345123456\ldots$$

each time concatenating the natural numbers from 1 to n and then continuing with 1 to $n+1$. Frida then asked Skye: 'How many numbers are you going to write?'. To which he replied, 'Just 2026 numbers'.

Task 2: Determine which the number is in the 2026th position.

Peter, who came late to the meeting, saw what they were doing and also wanted to share a problem with them: either using numbers beyond the naturals (like negatives or rationals), numbers arranged in a different way other than a line, or anything interesting he can think of.

Task 2: Help Peter create a nice problem he can give to Frida and Skye.